Project Planning & Management Document

Project Proposal, Gantt Timeline, Collaborative Task Assignment, Risk Mitigation & KPIs

Project Field	Project Specification Details
Project Title	EduHub — Full-Stack Online Learning Platform
Initiative & Program	Digital Egypt Pioneers Initiative (DEPI) — Software Engineering Track
Document Ref / Code	EDU-DOC-01 / DEPI-MS1
Document Version / State	Version 1.0 (Official Milestone 1 Submission)
Submission Due Date	16 October 2026
Architecture & Stack	N-Tier Architecture | ASP.NET Core MVC, EF Core, SQL Server, Azure, Stripe
Project Team Members	1. Aya Abdelmoniem Mohamed Hamed 2. Ibrahim Mohamed Ibrahim Abdo 3. Reem Ashraf Mohammed Ahmed 4. Ahmed Ramadan Ali Mohamed 5. Warda Mahmoud Khalaf Ahmed
Team Operating Model	Cross-Functional Collaborative Team (Collective ownership across all components; all members actively participate in full-stack implementation, architecture, DB, and QA)

1. Project Proposal — Overview, Objectives & Scope

1.1 Project Overview & Vision

EduHub is a full-stack online learning platform inspired by global leaders such as Udemy. Engineered using ASP.NET Core MVC, Entity Framework Core, and Microsoft SQL Server following a modular N-Tier Architecture, EduHub enables instructors to author, manage, and monetize structured courses while allowing students to discover courses, enroll seamlessly, watch cloud-delivered video lessons, complete interactive quizzes, track progress in real time, and earn verifiable PDF certificates.

Project Vision: To provide an accessible, user-friendly, secure, and high-performance online learning ecosystem that connects instructors with students and supports flexible, self-paced education without excessive platform commissions or infrastructure complexity.

1.2 Strategic Project Objectives

- **Structured Course Authoring:** Provide instructors with intuitive tools to create, organize (Sections and Lessons), and publish structured educational courses.

- **Course Discovery & Enrollment:** Enable students to search, filter, preview, purchase, and enroll in both free and paid courses.
- **Protected Cloud Video Streaming:** Integrate Microsoft Azure Blob Storage with time-limited Shared Access Signatures (SAS) and Video.js for secure, interactive video playback.
- **Progress Tracking & Assessments:** Accurately track lesson completion percentages on the server and evaluate student mastery through automated multiple-choice quizzes.
- **Automated PDF Certification:** Dynamically generate downloadable, tamper-resistant PDF certificates via iTextSharp once completion criteria are satisfied.
- **Secure Monetization & Discount Engine:** Process paid enrollments reliably via Stripe Checkout and verified webhooks, supporting flexible promotional coupons using the Strategy Pattern.
- **Platform Governance & Maintainability:** Empower administrators to moderate content, manage categories, and oversee users while adhering to clean N-Tier architecture and design patterns (Repository, Unit of Work, Strategy, Observer).

1.3 Project Scope Baseline (In-Scope, MVP & Out-of-Scope)

Scope Category	Included Modules & Functional Boundaries
In-Scope (Full Release)	1. ASP.NET Core Identity & Role-Based Authorization (Student, Instructor, Admin) 2. Course Discovery, Search, Multi-Filter & Category Navigation 3. Instructor Dashboard, Course Builder & Statistics 4. Azure Blob Storage Video Upload & Secure SAS Streaming 5. Student Enrollment & Server-Side Lesson Progress Tracking 6. Stripe Payment Processing, Webhook Verification & Coupon Discounts 7. Interactive Quizzes, Automated Grading & PDF Certificate Generation 8. Course Ratings/Reviews, Bookmarks, Lesson Notes, Announcements & Q&A
Minimum Viable Product (MVP)	Prioritizes the end-to-end core workflow: 1. Multi-role Registration, Login & RBAC 2. Course, Section & Lesson Creation by Instructors 3. Category Management & Course Review by Admins 4. Course Browsing, Free Enrollment & Stripe Paid Purchase 5. Video Lesson Playback & Progress Tracking 6. Quiz Completion & Downloadable PDF Certificate Issuance
Out-of-Scope (Initial Release)	Deferred to future post-graduation releases: - Live synchronous video classes (WebRTC) - Native iOS / Android mobile apps & offline video downloads - AI-based personalized recommendation engines - Multi-language UI localization & recurring subscription billing - Affiliate marketing & advanced BI data warehousing

2. Project Plan — Timeline, Gantt Chart, Milestones & Resource Allocation

2.1 Master Project Timeline & Visual Gantt Chart

The project schedule spans eight weeks from October 9, 2026 to December 4, 2026, structured into iterative Agile sprints directly synchronized with the four official Digital Egypt Pioneers Initiative (DEPI) submission milestones.

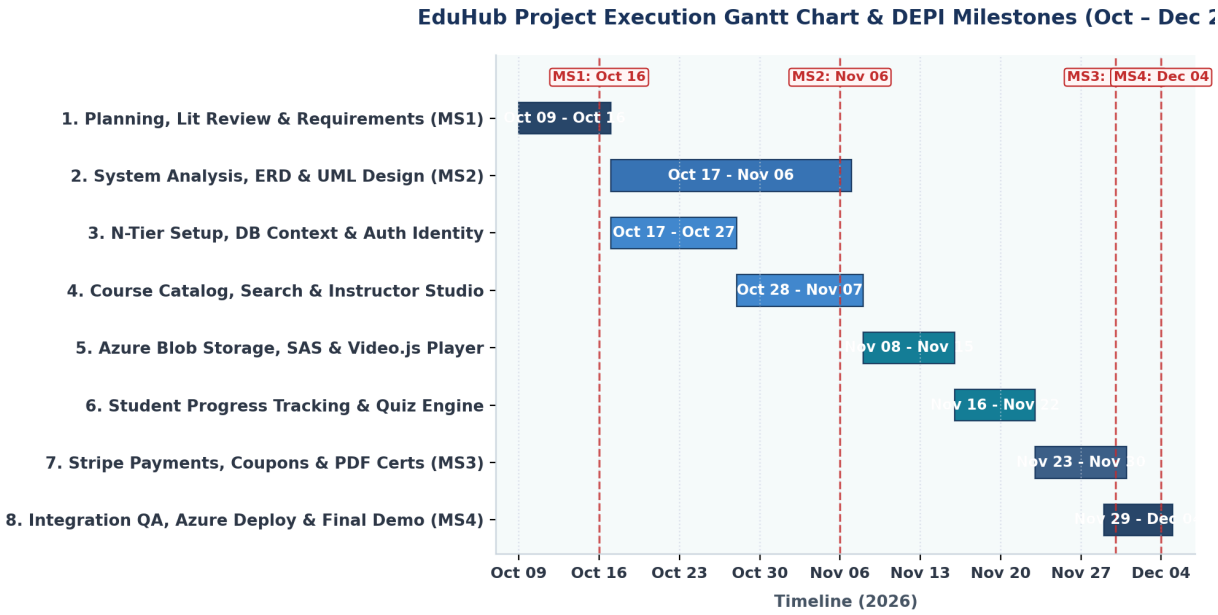


Figure 1: EduHub Master Project Gantt Chart & Official DEPI Milestone Schedule (Oct – Dec 2026)

2.2 DEPI Official Milestones & Key Deliverables

Milestone	Submission Date	Required Deliverables	Acceptance Gate Criteria
Milestone 1: Initiation & Requirements	16 Oct 2026	1. Project Planning & Management 2. Literature Review 3. Requirements Gathering	Approved scope, WBS, Gantt chart, benchmark study, and full FR/NFR catalog.
Milestone 2: System Analysis & Design	06 Nov 2026	1. Entity-Relationship Diagram (ERD) 2. UML Class, Use Case & Sequence Diagrams 3. Architecture & UI Wireframes	Normalized SQL schema, validated N-Tier contracts, and complete design blueprints.

Milestone 3: Core Implementation	30 Nov 2026	1. Complete N-Tier Source Code (GitHub) 2. EF Core Migrations & Seed Data 3. Working Azure & Stripe Integrations	Solution builds cleanly; all MVP workflows (Auth, Courses, Video, Stripe, Certs) operational.
Milestone 4: Testing & Final Presentation	04 Dec 2026	1. Unit, Integration & Security Test Reports 2. Deployed Azure Web Application 3. Final Presentation Deck & Live Demo	All E2E test scenarios pass; zero critical security flaws; successful live defense.

2.3 Phased Development Plan & Sprint Deliverables

Phase / Sprint	Dates	Technical Work Packages	Concrete Deliverables
Phase 1: Planning & Spec	Oct 09 – Oct 16	Charter, Scope, Lit Review, FR/NFR Elicitation, User Stories, RTM.	3 Official Milestone 1 Documentation Files.
Phase 2: Analysis & Setup (Development Week1)	Oct 17 – Nov 06	N-Tier Solution (Web, Services, Data, Tests), ERD, EF Core Context & Migrations, Repository & Unit of Work, ASP.NET Core Identity & Roles, Course Catalog & Search.	Working N-Tier architecture, migrated SQL DB, multi-role Auth, Course Browsing UI, and Milestone 2 Design Document.
Phase 3A: Instructor & Media (Development Week 2)	Nov 07 – Nov 18	Instructor Dashboard, Course Builder (Sections/Lessons ordering), Draft/Publish validation, Azure Blob Storage upload, Secure SAS token video access.	Working Instructor Studio, Course CRUD, Azure Blob upload flow, and protected Video.js preview.
Phase 3B: Student & Payments (Development Week3)	Nov 19 – Nov 30	Enrolled Course Portal, Lesson Progress Tracking, Quizzes & Grading, Stripe Checkout & Verified Webhooks, Strategy Pattern Coupons, iTextSharp PDF Certificates.	Complete Student Learning flow, Stripe Test Mode integration, Coupon engine, and automated PDF Certificates (Milestone 3).
Phase 4: QA & Handover	Dec 01 – Dec 04	Admin Moderation Workflows, Unit/Integration/Security Testing, Bug Fixing, Azure Cloud Deployment, Final Presentation.	Deployed production-ready platform, Test Report, and Final Defense Deck (Milestone 4).

2.4 Resource Allocation (Human, Technical & Cloud Resources)

Resource Type	Allocated Assets & Specifications	Utilization & Cost Governance
Human Engineering Resources	5 Full-Stack Software Engineers working ~20–25 hours/week each across an 8-week lifecycle (~800–1,000 total engineering person-hours).	100% shared allocation across Backend, Frontend, Database, Cloud, and QA.

Development Workstation & IDE	Visual Studio 2022 / VS Code, .NET 8 SDK, SQL Server Developer Edition / SSMS, Git & GitHub, Postman, Stripe CLI.	Free academic/community licenses; standardized across all 5 team machines.
Cloud & Storage Infrastructure	Microsoft Azure App Service (Hosting), Azure SQL Database, Azure Blob Storage (Private Media Containers) + Local Azurite Emulator.	Azure Student/Free Tier credits used for staging/production; Azurite used locally.
Third-Party APIs & Libraries	Stripe Payment Gateway (Test Mode Sandbox), Video.js (Open-Source HTML5 Player), Bootstrap 5, iTextSharp / QuestPDF, xUnit & Moq.	Zero licensing cost in test/academic mode; open-source MIT/LGPL packages.

3. Task Assignment & Roles — Collaborative Team Structure

> **Cross-Functional Collective Ownership Policy**
> All five team members work collaboratively across EVERY layer and module of EduHub (Presentation, Business Logic, Data Access, Cloud Integrations, and Testing). Rather than isolating members into narrow silos, every engineer contributes full-stack code across all features. To ensure smooth coordination and accountability, each member also serves as a Rotational Workstream Coordinator for specific modules while the entire team co-develops and peer-reviews the implementation.

3.1 Team Members & Full-Stack Role Allocation

Team Member Name	Shared Core Role	Primary Co-Leadership & Coordination Focus	Shared Full-Stack Responsibilities (All 5 Members)
Aya Abdelmoniem Mohamed Hamed	Full-Stack Software Engineer	Project Coordination, Database Architecture (ERD, EF Core, Migrations) & Course Discovery	• Write C# Controllers, Services & EF Repositories • Build Razor Views & Bootstrap 5 UI • Co-develop Azure & Stripe integrations • Write Unit/Integration Tests & review PRs
Ibrahim Mohamed Ibrahim Abdo	Full-Stack Software Engineer	System Architecture (N-Tier, Repository, Unit of Work), Azure Blob Storage & Video Streaming	• Write C# Controllers, Services & EF Repositories • Build Razor Views & Bootstrap 5 UI • Co-develop Azure & Stripe integrations • Write Unit/Integration Tests & review PRs

Reem Ashraf Mohammed Ahmed	Full-Stack Software Engineer	ASP.NET Core Identity, Role Authorization Policies, UI/UX Consistency & Quiz Engine	• Write C# Controllers, Services & EF Repositories • Build Razor Views & Bootstrap 5 UI • Co-develop Azure & Stripe integrations • Write Unit/Integration Tests & review PRs
Ahmed Ramadan Ali Mohamed	Full-Stack Software Engineer	Stripe Payment Gateway, Webhook Security, Strategy Pattern Coupons & Enrollment Logic	• Write C# Controllers, Services & EF Repositories • Build Razor Views & Bootstrap 5 UI • Co-develop Azure & Stripe integrations • Write Unit/Integration Tests & review PRs
Warda Mahmoud Khalaf Ahmed	Full-Stack Software Engineer	Instructor Dashboard, Progress Tracking, Observer Pattern, PDF Certificates & QA Testing	• Write C# Controllers, Services & EF Repositories • Build Razor Views & Bootstrap 5 UI • Co-develop Azure & Stripe integrations • Write Unit/Integration Tests & review PRs

3.2 Collaborative RACI & Task Distribution Matrix

In the matrix below, 'Lead (A/R)' denotes the member coordinating the task integration, while 'Co-Dev (R)' indicates active hands-on coding, pair programming, and pull-request validation by the rest of the team:

Workstream / Deliverable Module	Aya	Ibrahim	Reem	Ahmed	Warda
1. Project Planning, Lit Review & Requirements (MS1)	Lead (A/R)	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)
2. N-Tier Solution Setup, Repository & Unit of Work	Co-Dev (R)	Lead (A/R)	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)
3. Database ERD, Entities, Fluent API & Migrations	Lead (A/R)	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)
4. ASP.NET Core Identity, Roles & Security Matrix	Co-Dev (R)	Co-Dev (R)	Lead (A/R)	Co-Dev (R)	Co-Dev (R)
5. Course Discovery, Search, Filtering & Categories	Lead (A/R)	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)
6. Instructor Dashboard & Course/Section/Lesson Builder	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)	Lead (A/R)

7. Azure Blob Storage Upload & SAS Video Delivery	Co-Dev (R)	Lead (A/R)	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)
8. Student Learning Portal, Video.js & Progress Tracking	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)	Lead (A/R)	Co-Dev (R)
9. Quizzes, Automated Grading & Completion Rules	Co-Dev (R)	Co-Dev (R)	Lead (A/R)	Co-Dev (R)	Co-Dev (R)
10. Stripe Checkout, Webhooks & Coupon Strategy Pattern	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)	Lead (A/R)	Co-Dev (R)
11. PDF Certificate Generation & Admin Moderation	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)	Co-Dev (R)	Lead (A/R)
12. Automated Testing, Azure Deployment & Final Demo	Co-Dev (R)	Lead (A/R)	Co-Dev (R)	Co-Dev (R)	Lead (A/R)

4. Risk Assessment & Mitigation Plan

A rigorous risk assessment was conducted to identify technical, architectural, security, and schedule hazards that could impact EduHub delivery, paired with actionable prevention and contingency protocols:

Risk ID	Identified Risk & Root Cause	Probability	Impact	Mitigation & Prevention Strategy	Contingency Plan (If Triggered)
RSK-01	Scope Creep Exceeding Schedule: Attempting to build optional features before core workflow is stable.	Medium	High	Strictly enforce MVP boundaries; defer live classes, notes, and bulk uploads until core 7 MVP items pass QA.	Freeze secondary features immediately and reallocate all 5 members to critical path tasks.
RSK-02	Conflicting EF Core Migrations: Multiple members modifying entities and generating migrations concurrently.	High	High	Coordinate schema changes in daily standups; designate one migration merge window after team PR review.	Reset local branch migration snapshot, consolidate entity changes in `develop`, and regenerate a single clean migration.
RSK-03	Unauthorized Video Access / Piracy: Exposing permanent public URLs to paid course videos.	Medium	Critical	Store videos in private Azure Blob containers; issue short-lived SAS URLs (15-min TTL) only after server-side enrollment verification.	Revoke compromised storage account keys immediately and rotate SAS signing credentials.
RSK-04	Duplicate or Unverified Payments: Granting enrollment on browser redirect or processing duplicate webhooks.	Medium	Critical	Never grant access on client redirect; verify Stripe webhook cryptographic signatures and enforce database idempotency keys.	Run automated reconciliation script against Stripe payment intent logs and lock unverified enrollments.

RSK-05	Incorrect Discount Calculations: Client-side price manipulation or stacked invalid coupons.	Medium	High	Centralize pricing in server-side Strategy Pattern (`IPricingStrategy`); validate expiration, active flag, and usage limits.	Disable faulty coupon codes via Admin panel and fall back to standard base course pricing.
RSK-06	Duplicate Certificate Issuance: Race conditions or retry events generating multiple certificates.	Low	Medium	Enforce a unique database index on `(StudentId, CourseId)` in the Certificate table and idempotent Observer handlers.	Deduplicate records keeping the earliest issued serial number and invalidate redundant PDF blobs.
RSK-07	Frontend/Backend Integration Delays: Views and services developed with mismatched ViewModels/DTOs.	Medium	High	Define C# Service Interfaces, ViewModels, and API contracts upfront before parallel UI and backend coding.	Conduct an immediate pair-programming integration session to align DTO mappings.
RSK-08	External Service / Cloud Outages: Azure Blob Storage or Stripe sandbox configuration issues.	Low	High	Abstract external integrations behind interfaces (`IStorageService`, `IPaymentService`) and document test credentials.	Switch DI registration to Local Azurite Storage Emulator and Mock Payment Provider for uninterrupted local development.

5. Key Performance Indicators (KPIs) — Metrics for Project Success

To objectively evaluate the technical excellence, operational reliability, user experience, and engineering execution of EduHub, the team tracks concrete Key Performance Indicators (KPIs) across four dimensions:

KPI Category	Key Performance Indicator (KPI)	Target Benchmark / Threshold	Measurement & Verification Method
1. System Performance & Latency	Page Load & HTTP Response Time	< 1.5 seconds (95th percentile) for standard MVC views; < 300 ms for JSON API calls.	Measured via browser DevTools Network profiling and ASP.NET Core diagnostic middleware under 100 concurrent virtual users.
1. System Performance & Latency	Video Time-to-First-Frame (TTFB)	< 1.8 seconds initial video playback start via Azure Blob SAS streaming in Video.js.	Timed during integration testing across standard broadband connections for 720p/1080p MP4 lessons.
1. System Performance & Latency	PDF Certificate Generation Speed	< 2.0 seconds to verify completion, render PDF via iTextSharp, and return download stream.	Execution timer logged inside `CertificateService.GenerateCertificateAsync`.

2. Availability & Reliability	System Uptime & Availability	>= 99.5% uptime during staging and evaluation window on Microsoft Azure App Service.	Monitored via Azure App Service Health Check endpoints (`/health`).
2. Availability & Reliability	Payment & Webhook Accuracy Rate	100% transaction accuracy; 0% unverified enrollments; 0% duplicate webhook fulfillment.	Verified through Stripe CLI webhook replay tests and database transaction audit logs.
2. Availability & Reliability	Database Transaction Integrity	100% ACID rollback compliance on multi-entity failures via Unit of Work pattern.	Validated via automated integration tests simulating mid-transaction exceptions.
3. User Adoption & Learning UX	Core Workflow Completion Rate (Usability)	>= 95% task success rate for test users completing Registration -> Enrollment -> Lesson -> Quiz -> Certificate.	Measured during peer user-acceptance testing (UAT) across Scenarios A, B, and C.
3. User Adoption & Learning UX	Responsive UI & Accessibility Score	>= 90% Google Lighthouse Accessibility, Best Practices, and SEO score across Desktop & Mobile.	Audited using Chrome Lighthouse across Catalog, Instructor Studio, and Video Player views.
4. Engineering & Agile Delivery	Sprint Milestone On-Time Delivery	100% on-time submission across all 4 DEPI dates (Oct 16, Nov 6, Nov 30, Dec 4).	Tracked weekly against the project Gantt chart and GitHub Milestone burn-up boards.
4. Engineering & Agile Delivery	Automated Test & Security Pass Rate	100% pass rate on Unit/Integration test suite; 0 critical security defects (RBAC/IDOR/CSRF).	Verified on every GitHub Pull Request and pre-release security testing checklist (Scenario D).

5.1 Definition of Done (DoD) & GitHub Quality Gates

Every feature merged into the shared `develop` and `main` branches must satisfy all KPI thresholds and the team's Definition of Done:

(1) All acceptance criteria satisfied.

(2) N-Tier separation respected with zero business logic in Controllers.

(3) Resource-level authorization and server-side validation enforced.

(4) Unit and integration tests passing.

(5) Code reviewed and approved via Pull Request without secrets or connection strings committed.

(6) Documentation kept current.